

DSA Tutorial - 1

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Topic: String Operations

Div: SY1-B1

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Input:

```
#include <stdio.h>
```

```
int pallindrome(char input[]);
```

```
void copy(char input[], char output[]);
```

```
int stringlength(char input[]);
```

```
void substring(char input[]);
```

```
void reverse(char input[]);
```

```
int main() {
```

```
    int ch;
```

```
    char str[50];
```

```
    char output[50];
```

```
    printf("Enter a string: ");
```

```
    scanf("%49s", str);
```

```
    printf("Choose an Operation:\n1. Check Palindrome\n2. Copy String\n3. Check String\nLength\n4. Substring\n5. Reverse String\n");
```

```
    scanf("%d", &ch);
```

```
    switch(ch) {
```

```
        case 1: {
```

```
            int c = pallindrome(str);
```

```
            if (c == 1)
```

```
                printf("Palindrome\n");
```

```
            else
```

```
                printf("Not a Palindrome\n");
```

```
            break;
```

```
        }
```

```
        case 2:
```

```
            copy(str, output);
```

```
            break;
```

```
        case 3:
```

```
            printf("String length is: %d\n", stringlength(str));
```

```
            break;
```

```
        case 4:
```

```
            substring(str);
```

```

        break;
    case 5:
        reverse(str);
        break;
    default:
        printf("Invalid choice\n");
    }
    return 0;
}

```

```

int stringlength(char input[]) {
    int i = 0;
    while(input[i] != '\0') {
        i++;
    }
    return i;
}

```

```

int pallindrome(char input[]) {
    int i = 0, j = stringlength(input) - 1;

    while(i < j) {
        if(input[i] != input[j]) {
            return 0;
        }
        i++;
        j--;
    }
    return 1;
}

```

```

void copy(char input[], char output[]) {
    int i;
    for (i = 0; input[i] != '\0'; i++) {
        output[i] = input[i];
    }
    output[i] = '\0';
    printf("Output string is: %s\n", output);
}

```

```

void substring(char input[]) {
    int i, position;
    char output[50];
    int n = stringlength(input);
}

```

```

printf("Enter the position of substring: ");
scanf("%d", &position);

if(position >= 0 && position < n) {
    for(i = 0; i < n - position; i++) {
        output[i] = input[position + i];
    }
    output[i] = '\0';
    printf("Input string: %s\nOutput substring: %s\n", input, output);
} else {
    printf("Position entered is out of range\n");
}
}

void reverse(char input[]) {
    int i;
    int n = strlen(input);
    char output[50];

    for(i = 0; i < n; i++) {
        output[n - 1 - i] = input[i];
    }
    output[n] = '\0';
    printf("Input string: %s\nOutput (reversed) String: %s\n", input, output);
}

```

Output:

Output

```
Enter a string: harshB25et1038
```

```
Choose a Operation:
```

1. Check Pallindrome
2. Copy String
3. Check String lenght
4. Substring
5. Reverse String

```
3
```

```
String length is: 14
```

```
=== Code Execution Successful ===
```